POKHARA UNIVERSITY

Level: Bachelor Semester:Fall Year : 2020
Programme:B E Full Marks: 100
Course: Microprocessors Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Differentiate between Von Neumann and Harvard architecture. Explain fetch, decode and execute cycle in microprocessor with block diagram. 8
   b) What is the importance of DAA instruction? Explain it with an appropriate example along with the necessary calculations. 7
2. a) Write an ALP for 8085 to count the integers available in an array starting from 7 memory location 3500H to 3510H for exactly divisible by two and save the result in the register B. 7
   b) What is the significance of timing diagram? Draw and Explain a well labelled timing diagram of the instruction XRA B and calculate the total execution time if clock frequency is 2 mhz. 8
3. a) What is the use of 8255 PPI? Draw and Explain the control word format of 8255 PPI in I/O mode. 8
   b) Define interrupts. Differentiate between I/O mapped I/O and Memory mapped 7 I/O. 7
4. a) Draw a circuit for 8085 to interface 2KB RAM and 4KB ROM 8
   b) What is Interrupt Vector Table (IVT)? Draw the IVT for 8086 microprocessor 7 and explain different types of 8086 interrupts with respect to IVT. 7
5. a) Discuss the advantage of 8086 over 8085 and explain the concept of pipelining and segmentation 8
   b) Write an assembly language program in 8086 to find the largest number among 10 blocks of data and store the largest value in location "largest"
6. a) What is Macro assembler? Differentiate between Macros and Procedure.
   b) What do you understand by Null Modem? Show the interface of DTE and DCE with RS-232C.
7. Write short notes on: **(Any two)** 2
   a) DAM
   b) Flags of 8085
   c) Op codes, mnemonics and operands with example.

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